Project Overview

Introduction

The project component of Milestone II is designed to provide a comprehensive synthesis and assessment of the core data analytics skills that you've gained so far, including, for example, an end-to-end machine learning process applied to a challenging prediction problem. It's an opportunity to pursue all the steps in a typical end-to-end solution path for an interesting problem, including: analysis, implementation, evaluation, and communication of the problem and results, applied to more advanced/specialized problems similar to those you may encounter later in the degree, and in your career.

Project Support and Communication

Project Coach

You will be assigned a "project coach" from among the lead instructional faculty for this course. Your project coach will follow your project throughout its lifecycle in this course: from initial project proposal, to written project feedback, to the eventual grading of the written report at the end of the course. You are also required to have a discussion about the state of your project at least twice (by designated times specified in the syllabus) with your project coach. These meetings will happen asynchronously via DM in Slack, as well as synchronously via teams signing up for project meeting slots their coach makes available.

Project Team Slack Channel

Once teams have been formed, you will be added to a private Slack channel with your team members and project coach. Please use this channel to initiate project-related conversations with your coach and to schedule your meetings with your coach.

Standups

A rt of the course, you and your team will do 2 standup video meetings, as well as respond to other groups' standups. See the course syllabus for the dates on which these should happen. These will take place in a separate channel set up for the purpose. For more information on standups and what we're looking for in them, see this write-up on "How to do a Standup".

Due to the time-dependent nature of this assignment and the small point value, no partial credit will be given for late submissions of standup videos or peer feedback.

Project Overview and Policies

Dataset

For this project, you may choose your own dataset. You may use the same dataset for both the supervised and unsupervised portions of the project (see Methodology below); you may also use different datasets for each portion.

Please see the Dataset Guidelines page for the full set of data requirements. Any dataset you use must adhere to these conditions, or your project may not be approved.

Methodology

The project must incorporate both supervised and unsupervised learning components. You may use the same dataset for both parts or you may select different datasets for each.

1. Supervised learning component. Using your selected dataset, you will endeavor to use several different learning frameworks and feature representations. The goal will be to showcase your understanding of various supervised learning approaches including comparison and evaluation of model results.
2. Unsupervised learning. The goal here is to cluster the items in the dataset, or otherwise find an interesting structure, without labeled data, and to evaluate and describe your outcomes.

Project Development Components

The Milestone II project timeline will generally follow a similar one to that of Milestone I:

- Team formation and pre-proposal (via Google Sheet)
- Draft proposal (via Google Docs)
- Peer review of proposals (commenting on Google Docs)
- Asynchronous discussions with project coach (see above)
- Instructor-reviewed proposals (via Course)
- Synchronous discussion with project coach (see above)
- Final report (via Canvas Course and Google Docs)

Team formation: For Milestone II, we require by default that you work in **teams of three**. Although this recommendation can be relaxed to permit individual or two-person projects in exceptional circumstances, it has been our experience that working in teams balances high-quality work with the amount of work each person needs to do to complete the project. An example of one of the few scenarios where a one-person project would be allowed would be if that project required highly specialized domain expertise that would be problematic for other team members to learn quickly.

Project teams will receive one grade for submitted projects, which will be assigned to each member of the team. Grades are not assigned on an individual basis for team projects.

The rubric used to grade this Milestone project is the same regardless of team size: all teams must respond to the rubric's requirements as described here and have all required sections in their final report.

Overall, the goal of the Milestone II project work is to provide an opportunity to really bring together the diverse skills required of a data scientist, on problems that demonstrate your ability to apply the skills and concepts that you learned in the previous courses, while also being able to work in a domain and real dataset of your choice. In any case, for this project work you must make significant use of programming tools covered in prior MADS courses.

Project Proposal: Before starting work on the actual project, you'll write a short project proposal (see guidelines below) that is meant to be a high-level summary and does not need to contain technical details or code. You will discuss possible approaches to Part A (supervised learning) and Part B (unsupervised learning). You will then review your peers' proposals, and they will review yours. You can adjust your proposal based on input you receive and then you'll submit it to the course instructors for grading and approval. Requiring the proposal to be submitted early is intended to get you thinking about questions and datasets you're interested in, and how those might be answered with the tools you've been exposed to in previous classes.

Building on a Milestone I (or other previous course) Project

If you want to propose that your Milestone II project be a continuation of your Milestone I project (or any other course project), it must satisfy the dataset conditions above, and assuming it does, in your project proposal you must also provide a clear breakdown of the previous existing code you're re-using, compared to the new code you propose to create. We expect re-use to include code for data manipulation or integration: you cannot re-use any supervised or unsupervised learning code for which you got credit in Milestone I (or another course), as part of your Milestone II project. Failure to disclose such re-use may be considered academic misconduct.

Generative AI Policy

Your project's written report is the main way for you to communicate to us about your decision-making process and interpretation of your results. For this reason, the use of GenAI, such as ChatGPT, is prohibited for the written report. If these technologies are used for assistance in the coding portion of the project, the use of GenAI must be acknowledged or cited in the documentation and written report. As stated in the syllabus, students who are found to have used ChatGPT or the like outside of these requirements will be reported to UMSI Student Affairs for investigation as academic misconduct, and subject to consequences like failing the assignment or failing the course depending on the scope and severity of the actions taken.

Computing Choices

This course differs from your other MADS courses in many ways including technology. You have a number of choices described below. **Do not wait until the last minute to set up the computing environment you'll need!** You should have the computing environment you'll need for intensive training and evaluation ready <u>no later than week 5</u>, and preferably earlier, especially if you have special needs. If in doubt, ask the instructional team!

Additionally, **you will be required to submit all of your code files with your written report stored in a GitHub repository.** Additional assistance getting started with GitHub will be provided as needed.

1. We have created a Jupyter environment for you that is functionally equivalent to SIADS 516, which is a superset of the base MADS environment, and you can access that environment via the "ungraded lab assignment" in Course. This is a Vocareum "big instance" with 4 CPU and 16 Gb RAM. For file storage, you have a 1Gb limit as the max size for a single file, and you have 10Gb of total space. Use this environment if you have basic needs. If you plan to use deep learning that requires a GPU (which it usually does), then consider using one of the other options below, since we've had mixed results with the GPU support in Jupyter Labs.

You can use this Vocareum environment or choose to use any of the environments from courses you have already completed to build and test data manipulations and visualizations for your project. You can also use your own locally installed environment.

2. We encourage you to explore the use of cloud computing in your milestone project: you'll find it useful even if you're not doing deep learning, because the hyperparameter tuning and evaluation processes required by the report can also be computationally intensive. Also, it's good just to get experience using a new platform.

We strongly recommend making use of UMich cloud computing using the Great Lakes computing environment. This is especially true if you have very intensive computing needs (e.g. GPU for deep learning). Great Lakes is the University of Michigan's campus-wide computing cluster provided by ARC (Advanced Research Computing), a division of ITS (Information Technology Services). Great Lakes | ITS Advanced Research Computing. It is the shared, Linux-based high-performance computing (HPC) cluster available to researchers at the University of Michigan. Great Lakes consists of approximately 13,000 cores – including compute nodes composed of multiple CPU cores, GPUs, large-memory nodes, and support for simulation, modeling, machine learning, data science, genomics, and more.

All project based MADS courses offer this computing resource for students/teams with high end computing needs for their projects. Many previous Milestone 2 teams have used Great Lakes successfully. It is easy to set up, and you don't need to be an expert in Linux - in fact, there is a very straightforward Jupyter notebook mode!

3. Another possibility is to use Google Colaboratory or DeepNote, which may facilitate collaboration if needed. The School will not pay the access/subscription fees for these services.

Whatever computing environment you choose to utilize while working on your project, the files must ultimately be submitted by providing a link to a GitHub repository.